

I. NOTACIÓN DE DIRAC.

1) a) $|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$ ESTADOS POSIBLE COMPLETOS: $\langle\psi|\psi\rangle = 1$
 $|\alpha|^2 + |\beta|^2 = 1$

$\alpha = \frac{1}{\sqrt{2}} \quad \beta = \frac{1}{\sqrt{2}}$
 $\langle\psi|\psi\rangle = \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{1}{\sqrt{2}}\right|^2 = 1 \quad \checkmark$

$P(0) = \frac{1}{2} ; \quad P(1) = \frac{1}{2} ; \quad P(+)=\left(\frac{\langle 0| + \langle 1|}{\sqrt{2}}\right) \cdot \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) = \frac{1+1}{2} = 1$

$P(-) = \left(\frac{\langle 0| - \langle 1|}{\sqrt{2}}\right) \cdot \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) = \frac{1-1}{2} = 0$

b) $\psi = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle \Rightarrow \alpha = \frac{\sqrt{3}}{2} ; \beta = \frac{1}{2}$

$|\alpha|^2 + |\beta|^2 = \frac{3}{4} + \frac{1}{4} = 1 \quad \checkmark$

$P(0) = \left|\left(\langle 0| \cdot \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle\right)\right)\right|^2 = \frac{3}{4} ; \quad P(1) = \frac{1}{4}$

$P(+)=\left|\left(\frac{\langle 0| + \langle 1|}{\sqrt{2}}\right) \cdot \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle\right)\right|^2 ; \quad P(-)=\left|\left(\frac{\langle 0| - \langle 1|}{\sqrt{2}}\right) \cdot \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle\right)\right|^2$
 $= \left|\frac{\sqrt{3}}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}\right|^2 = \left|\frac{\sqrt{3}+1}{2\sqrt{2}}\right|^2 \approx 0,933$
 $= \left|\frac{\sqrt{3}-1}{2\sqrt{2}}\right|^2 \approx 0,067$

c) $|\psi\rangle = 0,7|0\rangle + 0,3|1\rangle \quad \alpha = 0,7 \quad \beta = 0,3$

$|\alpha|^2 + |\beta|^2 = (0,7)^2 + (0,3)^2 = 0,58 \quad \times \text{ NO ES ESTADO POSIBLE}$

$$d) 0,8^2 + 0,6^2 = 1 \quad \checkmark$$

$$P(0) = 0,64 \quad ; \quad P(1) = 0,36 \quad | \quad P(+)= 0,98 \quad | \quad P(-) = 0,02$$

$$e) \quad \cos(\theta) |0\rangle + i \sin(\theta) |1\rangle \quad | \alpha |^2 = \cos^2(\theta) = P(0)$$

$$\cos^2(\theta) + \sin^2(\theta) = 1 \quad | \beta |^2 = \sin^2(\theta) = P(1)$$

$$P(+)= \left| \frac{\cos(\theta) + i \sin(\theta)}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad P(-) = \frac{1}{2}$$

$$f) \quad |\alpha|^2 + |\beta|^2 = \cos^4(\theta) + \sin^4(\theta) \neq 1 \quad \times \quad \text{no é possível}$$

$$2) \quad |\Psi\rangle = \frac{3}{5} |0\rangle + \frac{4}{5} |1\rangle$$

$$\bullet \quad |0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}} \quad \bullet \quad |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}$$

$$\therefore |\Psi\rangle = \frac{3}{5} \left(\frac{|+\rangle + |-\rangle}{\sqrt{2}} \right) + \frac{4}{5} \left(\frac{|+\rangle - |-\rangle}{\sqrt{2}} \right) = \frac{7}{5\sqrt{2}} |+\rangle - \frac{1}{5\sqrt{2}} |-\rangle$$

II. COMPUERTAS LÓGICAS CUÁNTICAS DE UN QUBIT

$$1) \quad B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$$

$$a) \quad \bullet \quad B \text{ UNITARIA?} \quad \rightarrow \quad \underline{B \cdot B^t = I}$$

$$B^t = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \quad B \cdot B^t = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} =$$

$$= \frac{1}{2} \begin{pmatrix} 1 - i^2 & -i + i \\ i - i & -i^2 + 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \quad \checkmark$$

Handwritten signature

$$b) S = P(\pi/2) = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad H = \frac{(X+Z)}{\sqrt{2}}$$

$$U = H.S.B \quad H = \frac{1}{\sqrt{2}} \cdot \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H.S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$$

$$HSB = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1+i^2 & i+i \\ 1-i^2 & i-i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 2i \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}$$

$$2) a) M|0\rangle = (\alpha|0\rangle + \beta|1\rangle) \cdot \cancel{\langle 0|0\rangle} + (\beta^*|0\rangle - \alpha^*|1\rangle) \cdot \langle 1|0\rangle =$$

$$= \alpha|0\rangle + \beta|1\rangle \quad \checkmark$$

$$M|1\rangle = (\alpha|0\rangle + \beta|1\rangle) \cdot \langle 0|1\rangle + (\beta^*|0\rangle - \alpha^*|1\rangle) \cdot \langle 1|1\rangle =$$

$$= \beta^*|0\rangle - \alpha^*|1\rangle \quad \checkmark$$

$$b) M = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} \beta^* \\ -\alpha^* \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ \beta & 0 \end{pmatrix} + \begin{pmatrix} 0 & \beta^* \\ 0 & -\alpha^* \end{pmatrix} = \begin{pmatrix} \alpha & \beta^* \\ \beta & -\alpha^* \end{pmatrix}$$

$$c) M^t M = I \quad \rightarrow \quad M^t = \begin{pmatrix} \alpha^* & \beta^* \\ \beta & -\alpha \end{pmatrix}$$

$$M^t M = \begin{pmatrix} \alpha^* & \beta^* \\ \beta & -\alpha \end{pmatrix} \begin{pmatrix} \alpha & \beta^* \\ \beta & -\alpha^* \end{pmatrix} = \begin{pmatrix} |\alpha|^2 + |\beta|^2 & \alpha^* \beta^* - \alpha^* \beta^* \\ \beta \alpha - \alpha \beta & |\beta|^2 + |\alpha|^2 \end{pmatrix}$$

$$|\alpha|^2 + |\beta|^2 = 1 \quad \therefore \quad M^t M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \quad \checkmark$$

$$3) \cdot W = U_X (H \otimes I) \quad \cdot H = \frac{(X+Z)}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$U_X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix} \quad H \otimes I = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & I \end{pmatrix}$$

$$W = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix} \cdot \begin{pmatrix} I & I \\ I & -I \end{pmatrix} \cdot \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ X & -X \end{pmatrix}$$

• ESTADOS DE BELL:

$$\begin{cases} |\phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \\ |\phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) \\ |\psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) \\ |\psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \end{cases}$$

en base BELL:

$$W|00\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}; \quad W|01\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}; \quad W|10\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}; \quad W|11\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

III. COMPUTAS LÓGICAS CUANTAS DE LOS QUBITS

1) ~~scribble~~

a) $X \otimes I = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$

b) $I \otimes X = \begin{pmatrix} X & 0 \\ 0 & X \end{pmatrix}$

c) $X \otimes X = \begin{pmatrix} 0 & X \\ X & 0 \end{pmatrix}$

d) $U_X = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} =$
 $= \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & X \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix}$

LA PORTA FRANCO

VERIFICAR $U_X \cdot U_X^\dagger = I$

$$U_X |00\rangle = |00\rangle, \quad U_X |01\rangle = |01\rangle; \quad U_X |11\rangle = |11\rangle; \quad U_X |10\rangle = |10\rangle$$

COMO $X \in \mathbb{I}$ SON UNITARIAS $\Rightarrow X \otimes I, I \otimes X, X \otimes X, U_X$ SON UNITARIAS

2)

$$U_X = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$$

$$U_S = U_X \bar{U}_X U_X$$

$$\bar{U}_X = I \otimes |0\rangle\langle 0| + X \otimes |1\rangle\langle 1|$$

$$U_S |ab\rangle = |ba\rangle$$

$$U_X = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix} \quad \bar{U}_X = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix}$$

$$U_S = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix}$$

$$U_S |00\rangle = |00\rangle$$

$$U_S |01\rangle = |10\rangle$$

$$U_S |10\rangle = |01\rangle$$

$$U_S |11\rangle = |11\rangle$$

CRUCIANDO:

